Royal Mail Group

Royal Mail Address V1 REST API

Technical User Guide

API Specification

This API specification details the requirements for integrating with the Royal Mail Address V1 REST API. It specifically covers how Address V1 REST API can be used by business customers to obtain a Delivery Point Suffix (DPS) for UK addresses and provides the technical information to build this integration.

12th Sep 2022 Version 0.1

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1 Document Control

1.1 Terms and Abbreviations

Term	Meaning
API	Application Program Interface
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol over TLS
IP	Internet Protocol
REST	Representational State Transfer
Swagger	Specification for defining RESTful web services
TLS	Transport Layer Security
URL	Uniform Resource Locator
XML	Extensible Markup Language

Table 1 – Terms and Abbreviations

1.2 Version History

Version	Date	Author	Notes

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0.1	12/09/2022	RMG	Initial draft.

Table 2 – Document Version History

2 Overview

The Royal Mail Address API exposes a REST service that allows third parties and Royal Mail applications to match and verify a UK address.

There are no costs to customers for using the Royal Mail Address API V1 services, however customers' development costs must be covered by the customer developing the solution. Royal Mail will not accept any responsibility for this development, implementation and testing costs.

Customers should address initial enquiries regarding development of systems for these purposes to their account handler.

3 Purpose

This document is to provide Royal Mail customers with guidelines and detailed specifications for integrating with the Address V1 RESTful web service.

The document details:

- The specification for the web service interface is for providing consumers the facility to match and verify UK addresses.
- This API exposes the following capability:
  - Obtain DPS for a given address
- Description of errors the API can return.
- Non-functional characteristics of the API including response times, service availability and security considerations.

This document is primarily intended to be read by developers and other technical roles involved with integrating customer systems with the Address V1 API. This document should be read in conjunction with the following artefacts which are available from the 'Royal Mail Address V1 (REST)' page on the Royal Mail API (Developer) Portal:

- Address V1 REST API Swagger Definition (Click on 'Open API' link to download)

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4 Introduction to Address V1 API

4.1 Overview

The Royal Mail Address API exposes a REST service that provides third parties and Royal Mail applications the ability to match and verify a UK address. The API accepts a standard UK address as input and can provide either detailed or specific (e.g. DPS) information for that address.

Note: Some of the operations in the API specified in the Swagger/YAML file might be restricted and hence access could be limited only to the DPS operation.

4.2 Interface Components

Please see Figure 1 below for a graphical representation of the interface between Royal Mail and customers for Address V1 API. This document covers what information is to be exchanged, how this information is structured and the means by which it is transferred.

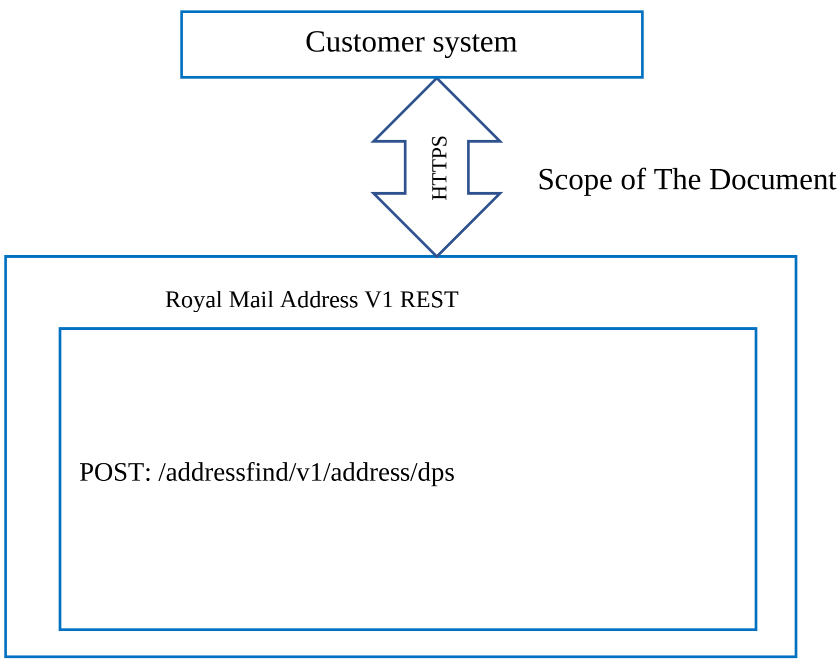

Figure 1 – Royal Mail Address V1 API

5 Integrating with Address V1 API

5.1 On-boarding Process

The high-level process associated with integrating Address V1 API is represented and described in the diagram below.

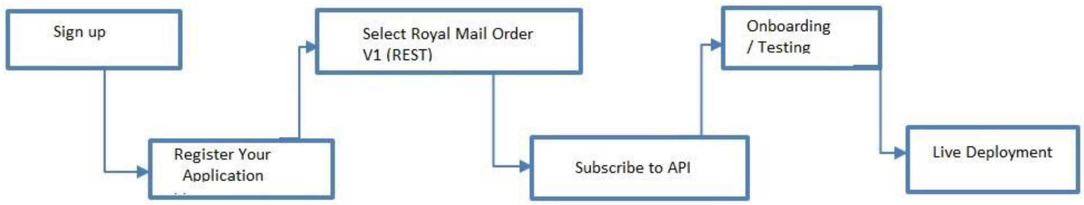

Figure 2 – Process for Integrating with the API

Access to the service is managed through RMG's API Management system.

New users of the system will need to:

1. Sign up for an account and accept the terms and conditions on the [Royal Mail API (Developer) Portal.]
2. Register the 'application' which will be calling the API. When the application is registered, it will be assigned a unique system-generated Client ID and Client Secret which is needed to securely access the API. It is important that these credentials are noted and securely stored.
3. Request to subscribe to the API. This will result in an e-mail being automatically generated and sent to the Royal Mail Customer Solutions team.
4. Once approved, testing can be performed against the API at controlled transaction rate.
5. Once all required testing/integration has completed, access to the Live API will be provided at a mutually agreed date/time.

Existing users who already have an account with Royal Mail's API Management system will need to perform step 2 onwards if the application accessing the API is different to any currently registered applications. If the application accessing the API is already registered, existing customers will need to perform step 3 onwards.

You will be provided with a contact in Royal Mail who will take you through the on-boarding process.

5.2 Terms & Conditions

You must accept the Royal Mail Terms and Conditions when creating your customer account. These cover the ways in which the service may be used and any integration activities must abide by these.

Of particular note to developers:

- The API is available 24x7 although must not be used for performance testing.
- The API imposes a cap on the number of transactions per second for each customer. Excessive volumes of traffic over the configured throttling limit will result in transactions being rejected.

- Royal Mail expects customers to use the service in a responsible way; this includes refraining from continuously polling the API for updates.
- Regular checks will be performed by Royal Mail to ensure that customers are using the API in a responsible way.
- Refer to **"Guide to populating Request data fields"** in the Appendix section of this document for additional info on composing the API request.

5.3 API Access

Access to the Address V1 API is via the following URL:

POST: https://api.royalmail.net/addressfind/v1/address/dps

The API only supports HTTP POST requests against above paths. Please note:

- Response formats supported: application/json
- See section 6 for all operations and parameters which are supported by this API.
- If anything is appended to the path and which is unsupported (e.g. /orders/something), then a HTTP 404 Not Found response will be returned.
- The Client must be provided in the HTTP header of all API requests otherwise access to the API will be rejected and a HTTP 401 (Unauthorised) will be returned. The Client ID is obtained by registering an application on the Royal Mail API (Developer) Portal.

5.4 API Versioning

Royal Mail is continuously working to improve its technology, and as part of this process updates to the services provided may on occasion necessitate a new API version. Royal Mail will look to maintain three versions of the API; as new versions are introduced; previous versions move down the stack until they are ultimately removed completely:

- Latest version
- Previous version
- Deprecated version

Customers will always be encouraged to integrate against the latest version as this will give them the longest stable period without the need to change, but if they have already begun integration activities when a new version is released then they will be able to integrate against the previous version. Customers should not integrate against the deprecated version.

6 Address Service

6.1 Business Services

The Address V1 API is a service offered to accept a standard UK address and in response provide a "Delivery Point Suffix" (DPS), a string of two characters that points to a specific delivery point when combined with a valid UK postcode. DPS is part of Royal Mail's PAF

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database and used by back-end systems and delivery staff. The table below provides an overview of the business services that are supported by this interface.

Business Service	API Operations	Description	Techno logy	Conversatio n Style
Find DPS	POST: /addressfind/v1/address/dps	Find DPS for the address provided in the request	JSON over HTTPS (REST)	Synchronous Request / Response

Table 3 – Business Services

6.2 HTTP Header Information

6.2.1 Description

The purpose of the HTTP header is to support security and logging functionally within the Royal Mail systems and it is mandatory that it is provided in the request message.

6.2.2 Request Headers

All service requests to this API will be authorise in accordance with the Client ID and Secret passed in the HTTP headers. Please see table below for the elements which need to be populated in the HTTP header.

Parameter	Optional	Description
X-IBM-Client-Id	No	Similar to a client username. Required to access the API.
X-IBM-ClientSecret	No	Similar to a client password. Required to access the API.
X-RMG-Date-Time	No	This should be populated with the date time in ISO 8601 format
X-RMG-Language	Yes	Optional default English
Accept	Yes	Content-Types that are acceptable for the response. Eg. application/json

Table 4 – HTTP Header Information in the API Request

6.2.3 Header Request Example Data

Example Request Data for the HTTP Header:

Parameter	Value
X-IBM-Client-Id	f0e***51-2041-4df2-b31d
X-IBM-Client-Secret	**0e***51-2041-4df2-b31d
X-RMG-Date-Time	2016-10-20T10:04:00+01:00
X-RMG-Language	en

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Accept	Application/json

Table 5 – Example HTTP Header Information for API Request

6.2.4 Response Headers

Please see table below for the elements which are populated in the HTTP response header.

Parameter	Optional	Description
HTTP/1.1	No	HTTP status code and reason phrase
X-BacksideTransport	No	OK OK on success or FAIL, FAIL on failure
Connection	No	Controls whether the network connection stays open after the current transaction finishes.
Content-Length	No	Indicates the content length
Content-Type	No	Indicates the media type of the resource.
Date	No	Contains the date and time at which the message was originated.
X-Global-Transaction-ID	No	Global transaction id
Access-Control-Expose-Headers	No	Indicates which headers can be exposed as part of the response by listing their names.
Access-ControlAllow-Origin	No	Indicates whether the response can be shared.
Access-ControlAllow-Methods	No	This response header specifies the method or methods allowed when accessing the resource.
X-RateLimitLimit	No	Request limit per day / per 5 minutes
X-RateLimitRemaining	No	The number of requests left for the time window

Table 6 – HTTP Header Information in the API Response

6.2.5 Header Response Example Data

Example Response Data for the HTTP Header:

Parameter	Value
HTTP/1.1	200 Ok
X-Backside-Transport	OK OK
Connection	Keep-Alive
Content-Length	4566

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Content-Type	application/json
Date	14 Feb 2020 14:33:16 GMT
X-Global-Transaction-ID	5s****fd8*****afacae13d469
Access-Control-Expose-Headers	APIm-Debug-Trans-Id
Access-Control-Allow-Origin	*.royalmail.com
Access-Control-Allow-Methods	POST
X-RateLimit-Limit	rate-limit-1,220;
X-RateLimit-Remaining	name=rate-limit-1,219;

Table 7 – Example HTTP Header Information for API Response

6.3 Find DPS Operation

6.3.1 Description

The behaviour of this operation is to return a DPS (Delivery Point Suffix) for the addresses provided in the API request (typically these are customer/consumer provided addresses). There is a limit on the number of address that can be batched together for efficiency. The limit is specified in the swagger/YAML for the API.

6.3.2 Request Message

URL Path POST: https://api.royalmail.net/addressfind/v1/address/dps

Please refer to section 6.2 for the request and response header information.

Parameter Name	Optional	Type	Description
address	No	Object	Address object contains the fields from a standard UK address. For more details check the example below (and refer to the swagger file). More information on populating the request fields is available in the appendix section **"Guide to populating Request data fields"**.

Table 8 – Request message to Find DPS operation

6.3.3 Response Message

A successful business response will be returned as a standard HTTP response code of 200.

Parameter Name	Optional	Type	Description
Input	No	Object	Contains the address object from the request message to be able to associate with the DPS field in the response.

dps	No	string	Delivery Point Suffix – string of max length 2. Typically, the first character is a number and the second is an alphabet.

Table 9 – Response message for Find DPS operation

6.3.4 Find DPS Operation Example Data

Full JSON example responses are provided on the Royal Mail API (Developer) Portal.

Example Request Data

```
POST https://api.royalmail.net/addressfind/v1/address/dps HTTP/1.1
Accept-Encoding: gzip,deflate
Content-Type: application/json
X-RMG-Date-Time: 2016-10-20T10:04:00+01:00
X-IBM-Client-ID: 00*****1-5**9-4**1-9**4-6**********0
X-IBM-Client-Secret: hU2goA3rK8bY0*******************oW
Content-Length: 916
Host: api.royalmail.net
Connection: Keep-Alive
User-Agent: Apache-HttpClient/4.1.1 (java 1.5)
```


```
{
  "Addresses": [
    {
      "addressLine1": "Flat 43, Berberis House",
      "addressLine2": "Highfield Road",
      "postTown": "Feltham",
      "County": "Middlesex",
      "postcode": "TW13 4GP"
    }
  ]
}
```


Example Response Data

```
HTTP/1.1 200
X-Backside-Transport: OK OK
Connection: Keep-Alive
Content-Length: 1232
Content-Type: application/json
Date: Mon, 24 Aug 2020 06:36:31 GMT
X-Global-Transaction-ID: 5521cfd85f435fef1345a631
Access-Control-Expose-Headers: APIm-Debug-Trans-Id, X-RateLimit-
Limit, X-RateLimit-Remaining, X-RateLimit-Reset, X-Global-
Transaction-ID
Access-Control-Allow-Origin: *
Access-Control-Allow-Methods: POST
X-RateLimit-Limit: name=rate-limit-1,120;
X-RateLimit-Remaining: name=rate-limit-1,119;
Set-Cookie: DPJSESSIONID=PBC5YS:464726495; Path=/;
Domain=.royalmail.net
```


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```
[
  {
    "Input": {
      "addressLine1": " Flat 43, Berberis House",
      "addressLine2": " Highfield Road",
      "addressLine3": "string",
      "postTown": " Feltham",
      "County": " Middlesex",
      "postcode": " TW13 4GP"
    },
    "DPS": "1F"
  }
]
```


7 Error Handling

7.1 Overview

There are two types of errors produced by the Address V1 API, namely:

- Business Errors (e.g. postcode invalid etc.)
- Technical Errors / Exceptions (e.g. service unavailable etc.). There are two types of technical errors:

  1. API Gateway Errors. Returned when the API request is rejected by the API Gateway before any business processing takes place. E.g. rate limit exceeded.

  2. Application Errors. Returned when API requests are accepted for business processing but processing cannot be completed due to a technical error. E.g. internal server error.

Both sets of errors should be appropriately handled by your system and technical details of the error should not be displayed directly to consumers. Please refer to the tables below for the generic structure of all core error messages returned by Address V1 API.

The tables below show the fields for API Gateway and non-API Gateway type error responses.

API Gateway Error

Field	Optional	Type	Description
httpCode	No	String	HTTP error code
httpMessage	No	String	HTTP error code description
moreInformation	Yes	String	Information relating to the error condition

Table 12 – API Gateway Error Response Structure

Non-API Gateway Error (Business & Application)

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Field	Optional	Type	Description
Errors	Yes	Array	Array containing error information
errors.code	Yes	String	Code associated with the error condition Yes
errors.description	Yes	String	Description of the error condition
Field	Optional	Type	Description
errors.cause	Yes	String	Cause of the error (if known)
errors.resolution	Yes	String	Description of the resolution and action required to correct the error

Table 13 – Non-API Gateway Error Response Structure

The following sections describe the content of the business and technical errors returned by Address V1 API with their respective HTTP status code response.

7.2 Business Errors

All errors associated with processing the API requests are returned using the error structure defined in section 7.1.

For missing or empty mandatory field's scenario:

HTTP Status Code	code	description	cause	resolution
400	E1398	[Field] is required and must not be empty	Missing or empty mandatory field	Check mandatory field(s) and resubmit
400	E1705	Number of addresses in the request array should be {count} or less	Invalid request parameters	Please check and re-submit

Table 14 – API Business Errors

7.2.1 Example Data

Please see below for an example of a business error which is returned which results in an E1329 error being returned. Full JSON example responses are provided on the [Royal Mail API (Developer) Portal](#).

```
HTTP/1.1 400
Content-Type: application/json
```


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```
{
  “httpCode”: “400”,
  “httpMessage”: “Bad Request”,
  "errors": [
    {
      "code": "E1398",
      "description": "[Field] is required and must not be empty",
      "cause": "Missing or empty mandatory field",
      "resolution": "Check mandatory field(s) and resubmit"
    }
  ]
}
```


7.3 Technical Errors / Exceptions

The following technical exceptions/error scenarios will be caught and handled as described in the tables below.

The tables below show API Gateway and Application technical errors. Only the most common API Gateway error descriptions are shown, with the HTTP status return code considered adequate for other cases and hence other response codes and descriptions are not listed here.

API Gateway Errors

HTTP Status Code	httpCode	httpMessage	moreInformation
401	401	Unauthorized	Invalid client id or secret
404	404	Not Found	API not found for requested URI
405	405	Method Not Allowed	The method is not allowed for the requested URL

Table 15 – API Gateway Technical Errors

Application Errors

HTTP Status Code	code	description	cause	resolution
500	E0000	Internal exception occurred	An internal error was identified while attempting to process your API request	Please try again later

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503	E0001	Service unavailable	An internal error was identified while attempting to process your API request	Please try again later

Table 16 – Application Technical Errors

For all other technical issues please contact a Royal Mail Support representative by visiting the Royal Mail API (Developer) Portal Support pages.

7.3.1 Example Data

Please see below for examples of both API Gateway and NonGateway technical errors which are returned in the event that the rate limit for API calls has been exceeded or if an internal error is encountered whilst processing the request.
Full JSON example responses are provided on the Royal Mail API (Developer) Portal.

API Gateway Error

```
HTTP/1.1 429
Content-Type: application/json

{
 "httpCode": "429",
 "httpMessage": "Too Many Requests",
 "moreInformation": "The rate limit has been exceeded for the plan or operation being used."
}
```

Non-API Gateway Error

```
HTTP/1.1 500
Content-Type: application/json

{
 “httpCode”: “500”,
 “httpMessage”: “Internal server Error”,
 "errors": [
   {
    "code": "E0000",
    "description": "Internal exception occurred",
    "cause": "An internal error was identified while attempting to process your API request",
    "resolution": "Please try again later"
   }  ]
}
```

8 Non-Functional Characteristics

8.1 Availability

8.1.1 Service Hours

The Address API is available 24 hours per day x 365 days per year.

8.1.2 Maintenance Windows

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Royal Mail Online Services Terms and Conditions define the maintenance for this service.

8.1.3 Unavailability

In the unlikely event of the API being unavailable, customer systems should make provision to handle this appropriately.

If you experience issues with the availability of this API please contact a Royal Mail Support representative by visiting the Royal Mail API (Developer) Portal Support pages.

8.2 Performance

Royal Mail aims to respond to calls in less than 5 seconds on average when invoked from the edge of Royal Mail's UK data centre.

8.3 Security

The REST API will only accept requests and return responses over HTTPS. All service requests via the API Management solution will be authorised in accordance with the below security parameters passed in the HTTP headers. This will ensure that any external service requests are authorised and authenticated in line with RMG Security Policies and Standards.

1. Client Id and client secret

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9 Frequently Asked Questions

Please see the [FAQ page] on the [Royal Mail API (Developer) Portal] for a general list of frequently asked questions with responses.

All FAQs specific to the API described in this document are listed below.

9.1 Latest versions of Swagger Definition

Question: Where can I find the latest version of the Address V1 API Swagger definition?

Answer: The latest version of the Swagger definition can be found on the 'Royal Mail Address V1 (REST)' page on the Royal Mail API (Developer) Portal. The 'Open API' link can be used to download the swagger.

9.2 Sample Code

Question: Do Royal Mail provide any sample code for the Address V1 API to help accelerate my integration?

Answer: Yes, the Royal Mail API (Developer) Portal provides sample code for a number of programming languages including cURL, PHP, Ruby, Python, Java, Node, Go, Swift etc. Please navigate to the 'API Library' and select 'Royal Mail Address V1 REST API'. A variety of programming languages will be provided on the right hand-side of the page.

9.3 API Programming

Question: Can Royal Mail complete the API programming for me?

Answer: Royal Mail only provides user guides to enable an understanding of the API and therefore cannot complete any programming or system development for your business.

9.4 Business Account

Question: I don't have a business account with Royal Mail. Can I use the Address V1 API?

Answer: No – Address V1 API is only available to Royal Mail account holders.

9.5 Application Compatibility

Question: What Software Development Kits or tools have been proven to work with the Address V1 API?

Answer: The following applications are known to be compatible with the Royal Mail Address V1 API: SoapUI.

10 Appendix

a Guide to populating Request data fields

Tags	Description	Mandatory(Y/N)	Sample Values	Comments
Body				
addresses	Array containing one or more addresses for which the DPS is to be obtained.	Y		Although only few fields are mandatory in the address object (see rows below), populating all relevant content for a UK address (like door/house/building number, street, county, town/city, etc) will make the address matching more accurate and hence increase the accuracy of the DPS value returned. DO **NOT** INCLUDE DELIVERY INSTRUCTIONS (E.G. SAFE PLACE INSTRUCTIONS OR DELIVER TO NEIGHBOUR INSTRUCTIONS) IN ANY OF THE FIELDS BELOW. INCLUDING SUCH DATA AS PART OF ANY OF THE ADDRESS FIELDS REDUCES THE ACCURACY OF ADDRESS MATCHING AND HENCE CAN POTENTIALLY AFFECT THE ACCURACY OF THE DPS VALUE RETURNED BY THE API.
addressLine1	First line of address from a standard UK address.	Y	e.g. “Flat 43, Berberis House”	Since the max length of this field is limited to 60 characters, part of the string after 60th character should be cut and populated into addressLine2 (second line of address)

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addressLine2	Second line of address from a standard UK address.	N	“Highfield Road”	Since the max length of this field is limited to 60 characters, part of the string after 60th character should be cut and populated into addressLine3 (third line of address)
addressLine3	Third line of address from a standard UK address.	N		Since the max length of ths field is limited to 60 characters, part of the string after 60th character should be truncated.
postTown	Name of the town/city to which the address belongs.	Y	“Feltham”	Do not club multiple values (e.g. town name + county name, or town name + borough name, or town name + street name, etc.) for this field. Doing so will reduce the accuracy of the DPS returned by the API.
county	Name of the UK county.	N	“Middlesex”	Do not club multiple values (e.g., town name + county name, etc) in this field.
postcode	Standard UK postcode.	Y	“TW13 4GP”	UK postcode contains two parts: First part starts with one or two letters followed by one or two numeric digits. Second part starts with one numeric digit followed by two letters. Typically, there is white-space (blank character) present in between the two parts but is not mandatory. There is no need to pad the value of this field with leading or trailing whitespaces (blank characters). Populate both parts accurately for best results. Not providing a valid, complete postcode reduces the efficiency of the matching algorithm and hence can potentially

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				result in inaccurate DPS value being returned in the API response.

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